U23IT484 – CLOUD COMPUTING TECHNOLOGIES

Name: Santhu Mohammed M

Roll No: 24AM089

ROUND 2 (i) – IMPLEMENT AWS STORAGE

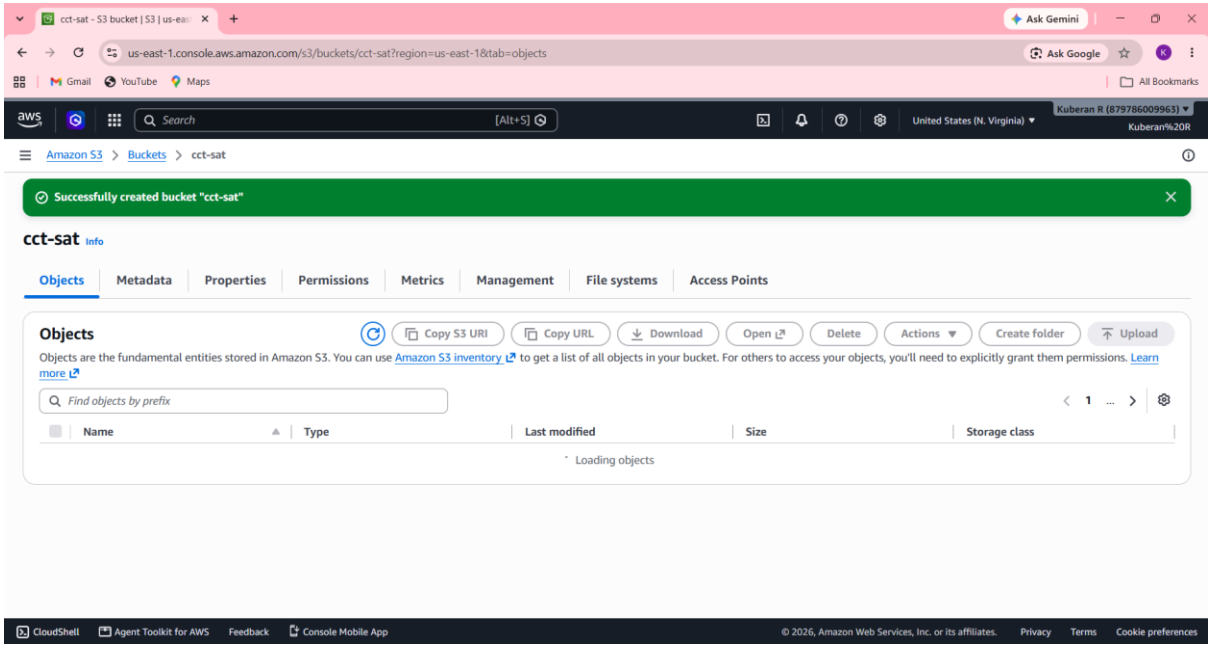
Objective

To implement secure cloud storage using **Amazon S3** with encryption, versioning, access control, and protection against accidental or unauthorized data deletion.

Procedure

Step 1 – Open Amazon S3

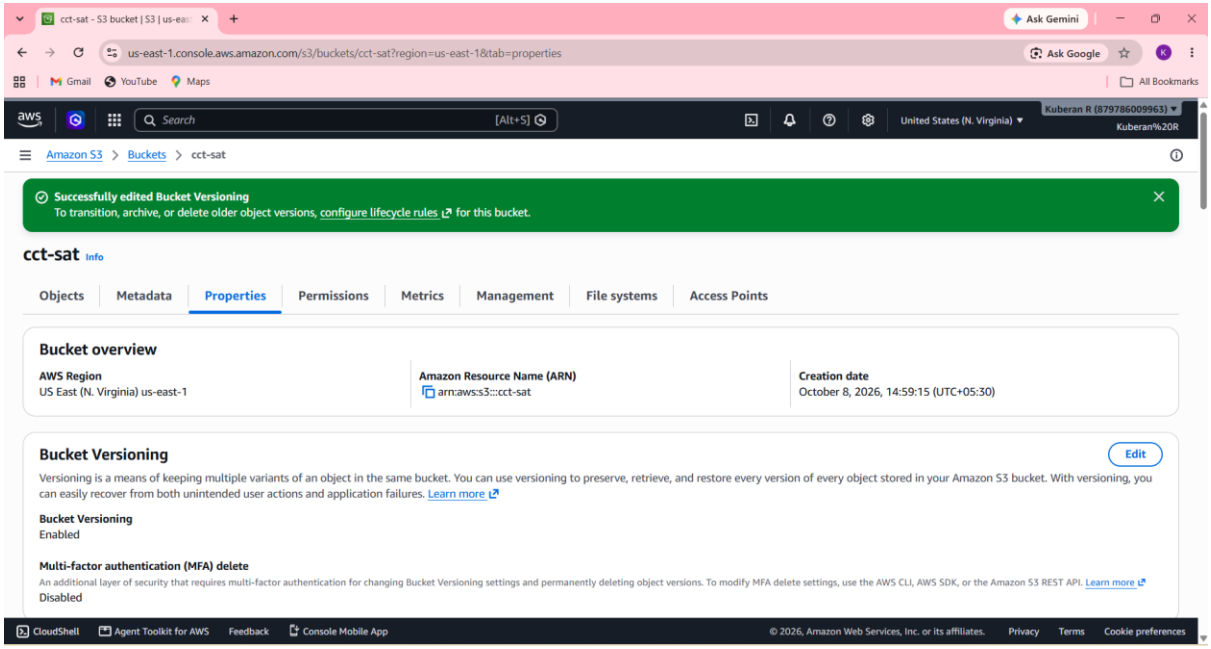
1. Sign in to the **AWS Management Console**.
2. Search for **S3** and open the Amazon S3 service.
3. Navigate to **Buckets** and select **Create bucket**.
4. Enter a globally unique bucket name.
5. Select the required AWS Region.
6. Keep **Block all public access** enabled.
7. Create the bucket.



Step 2 – Enable S3 Versioning

1. Open the newly created S3 bucket.
2. Go to the **Properties** tab.
3. Find **Bucket Versioning**.
4. Select **Enable**.
5. Save the configuration.

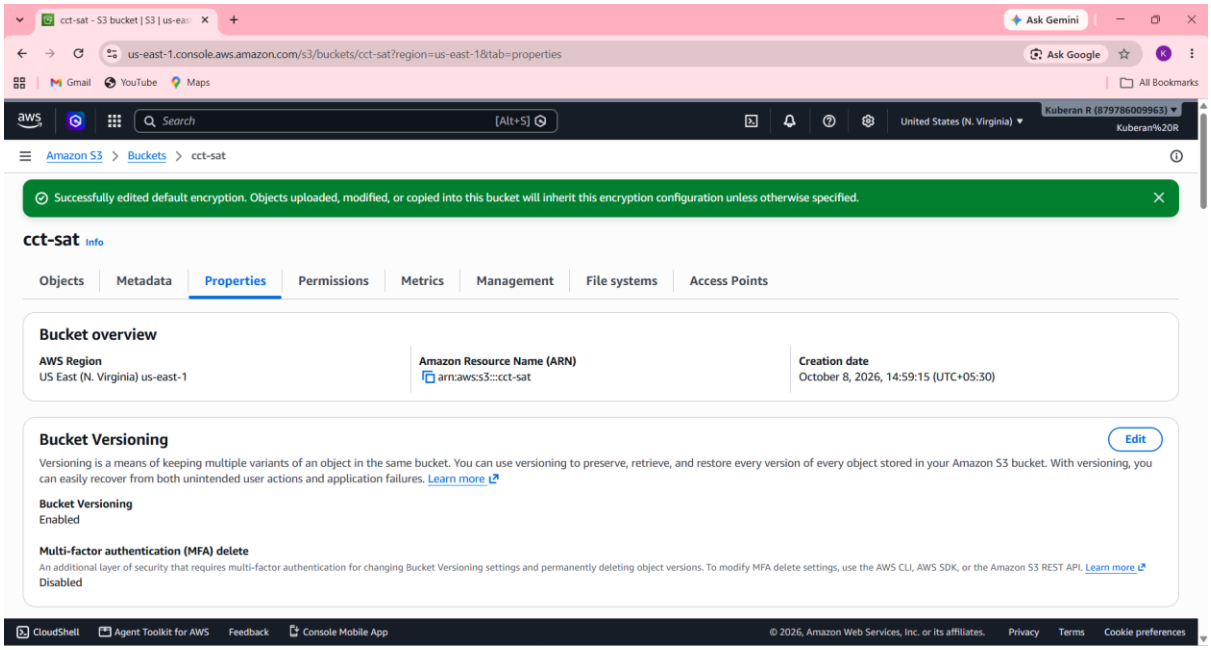
Versioning helps retain previous versions of objects and provides recovery from accidental overwrites or deletions.



Step 3 – Configure Server-Side Encryption

1. In the S3 bucket, open the **Properties** tab.
2. Locate **Default encryption**.
3. Enable server-side encryption.
4. Select **SSE-S3** for a simple student implementation.
5. Save the configuration.

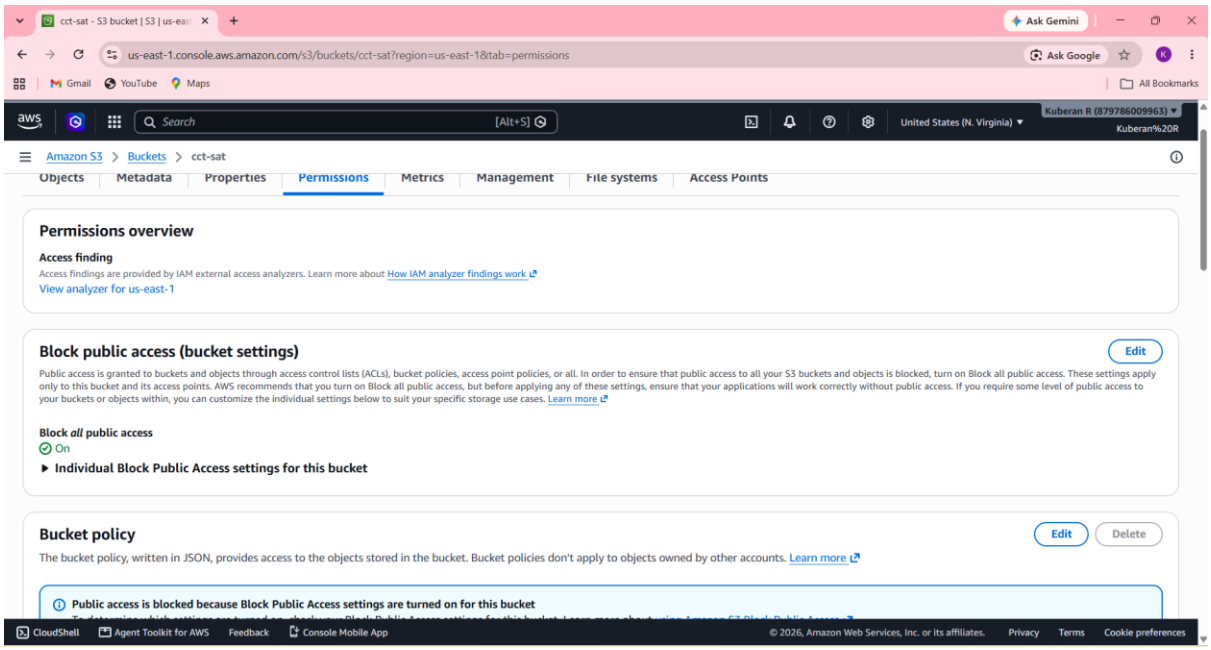
This ensures that objects stored in the bucket are encrypted at rest.



Step 4 – Configure Public Access Protection

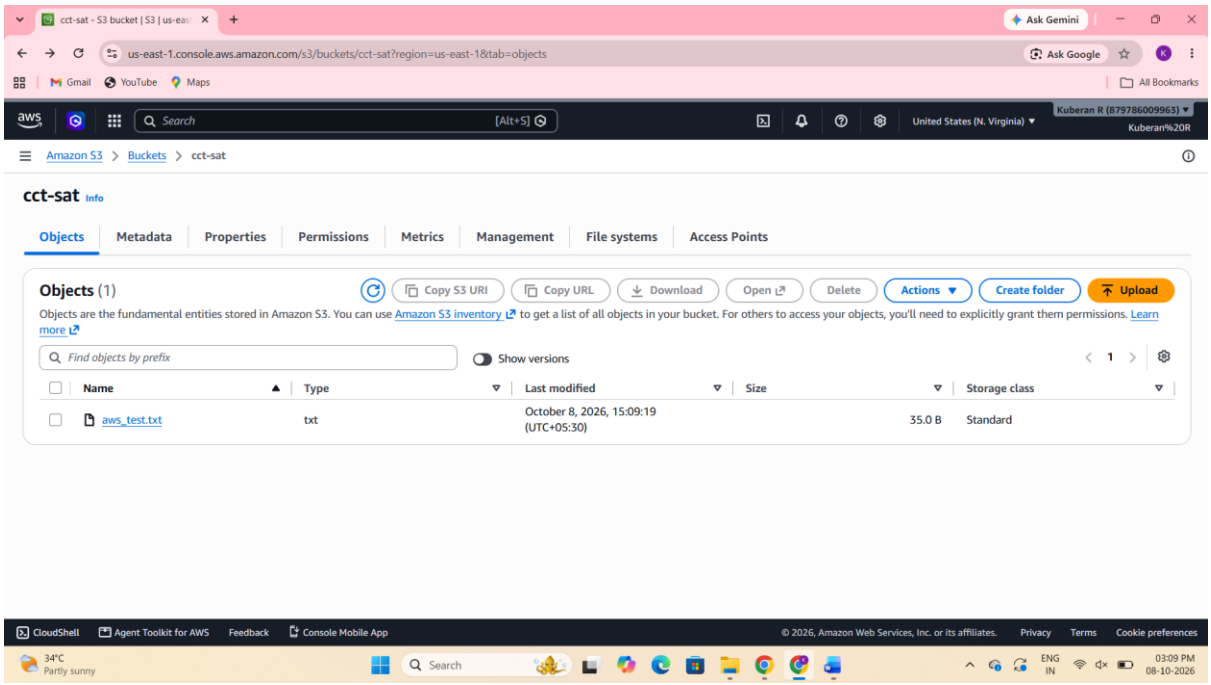
1. Open the bucket's **Permissions** tab.
2. Verify that **Block all public access** is enabled.
3. Confirm that no public bucket policy is configured.
4. Save the settings if any changes were made.

This prevents the storage bucket from accidentally becoming publicly accessible.



Step 5 – Upload a Test Object

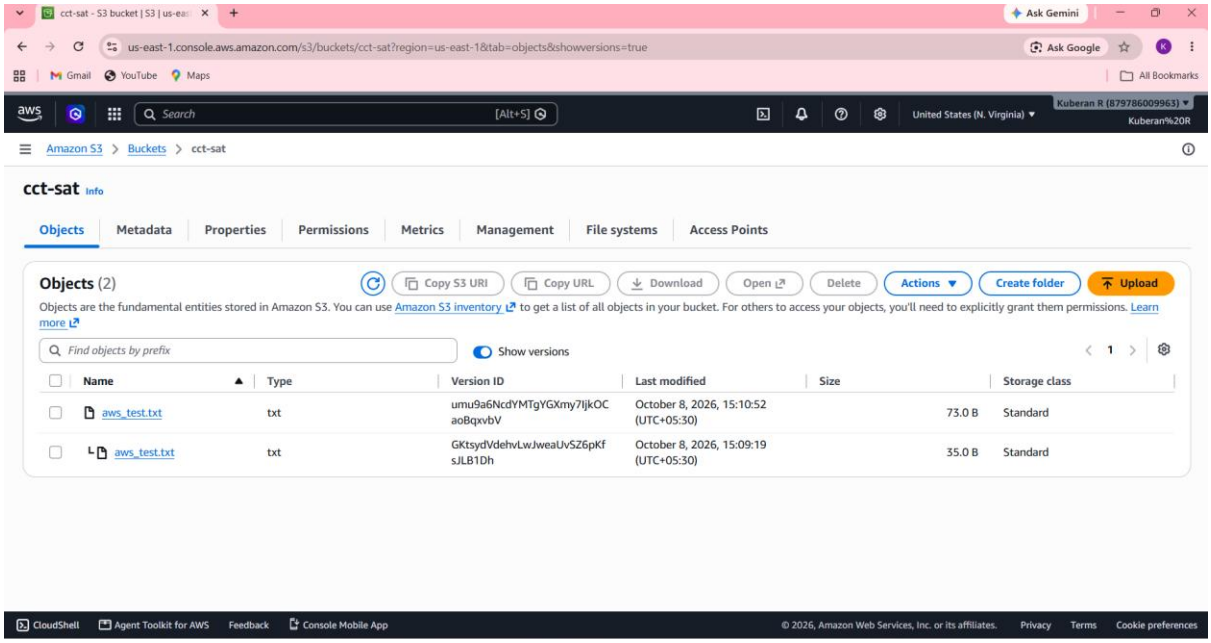
1. Open the S3 bucket.
2. Select **Objects → Upload**.
3. Upload a small test file.
4. Verify that the object appears in the bucket.
5. Open the object's Properties and verify its encryption/version information.



Step 7 – Test Versioning

1. Upload a file with the same name again after modifying its contents.
2. Open the bucket and enable **Show versions**.
3. Verify that multiple versions of the object are available.

This demonstrates how S3 versioning can help recover previous data.



ROUND 2 (ii) – BUILD END-TO-END AWS SOLUTION

Objective

To build an end-to-end AWS cloud solution by integrating **VPC, EC2, RDS, S3, IAM, and security controls** to provide a secure and functional application environment.

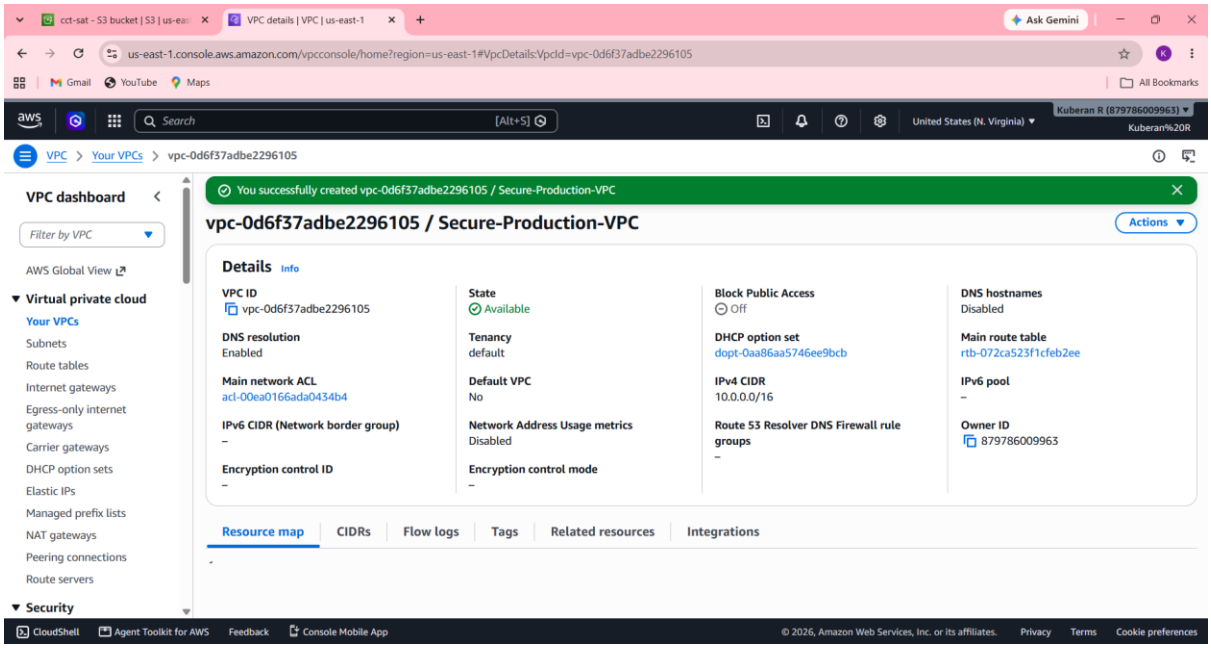
Procedure

Step 1 – Create the VPC

1. Open **AWS Console → VPC**.
2. Select **Create VPC**.
3. Choose **VPC only**.
4. Give the VPC a suitable name, such as Secure-Production-VPC.
5. Use a private CIDR range such as:

   10.0.0.0/16

6. Create the VPC.



Step 2 – Create the Subnets

Create separate subnets for the application and database components.

For example:

VPC: 10.0.0.0/16

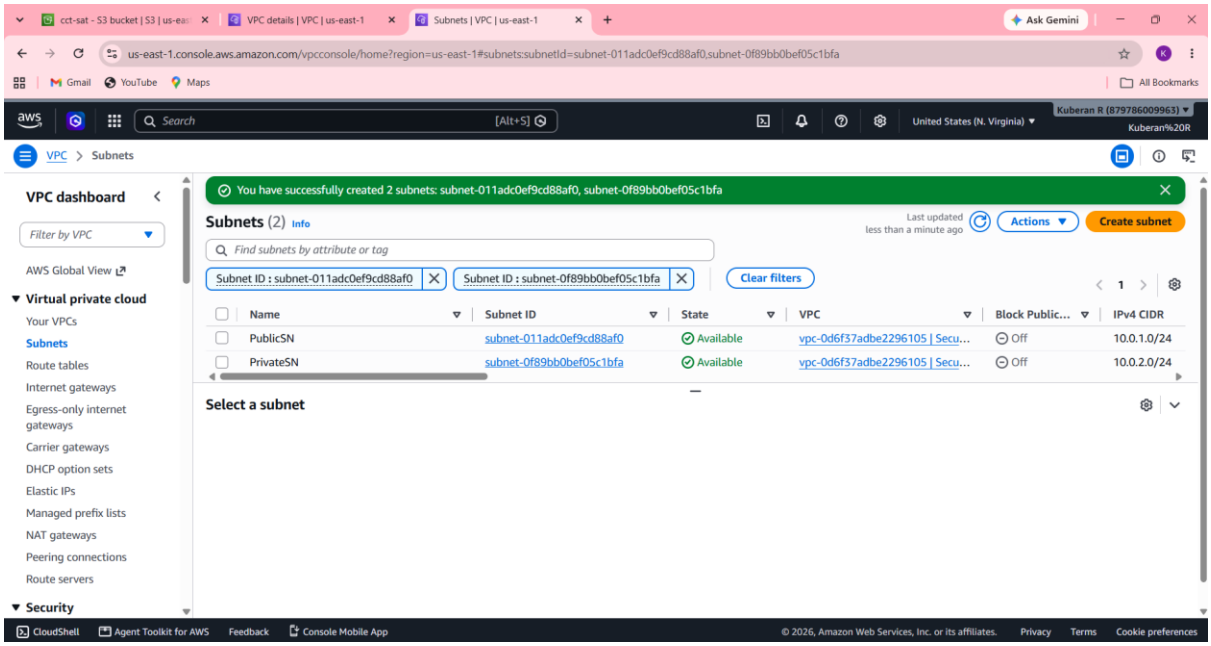
Public Subnet:

10.0.1.0/24

Private Subnet:

10.0.2.0/24

1. Open **VPC → Subnets**.
2. Create the public subnet.
3. Create the private subnet.
4. Associate both with the production VPC.



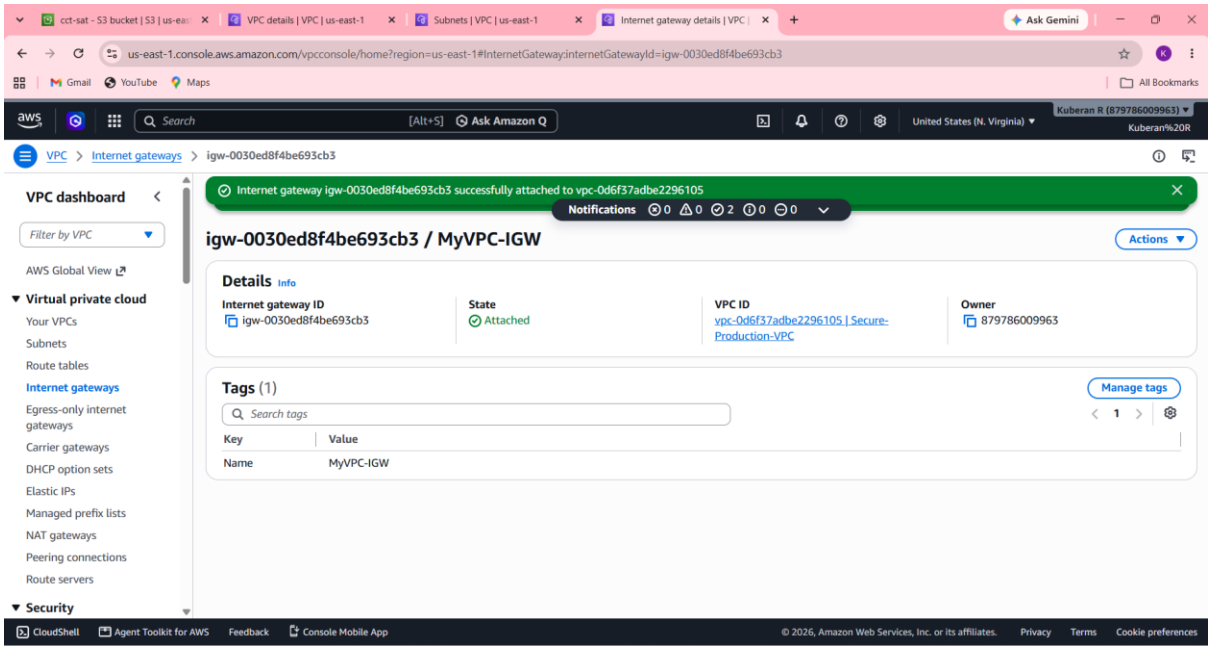
Step 3 – Configure Internet Connectivity

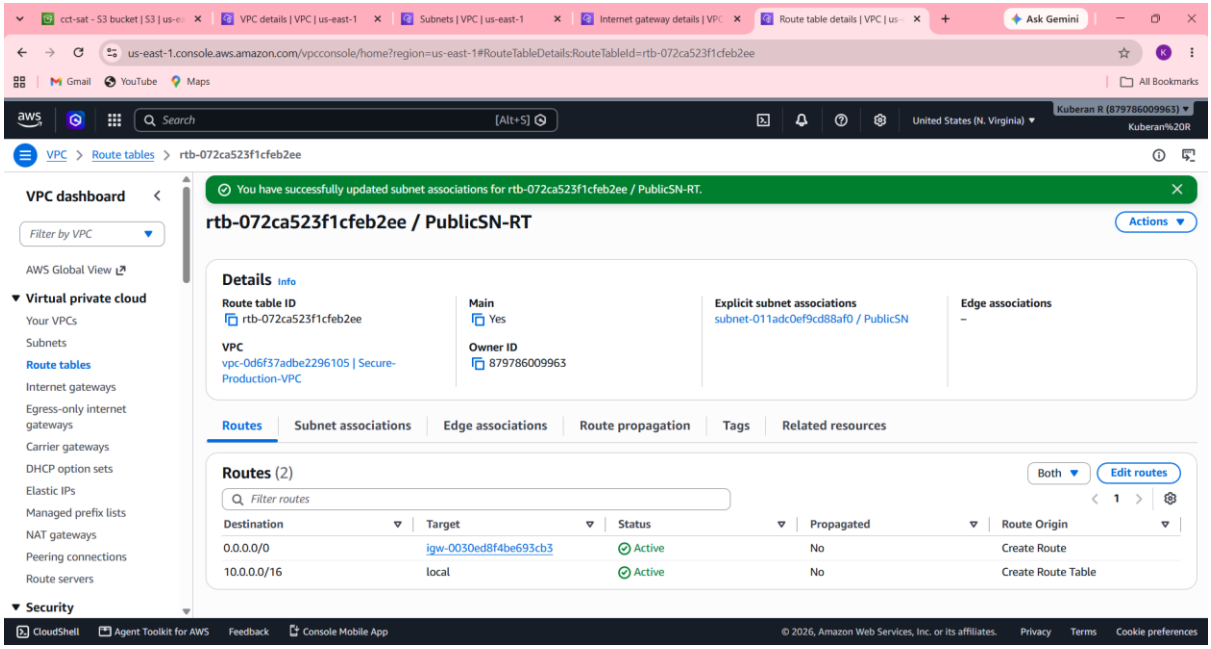
For the EC2 application server to receive required internet connectivity:

1. Create an **Internet Gateway**.
2. Attach it to the VPC.
3. Create/configure a route table for the public subnet.
4. Add the route:

0.0.0.0/0 → Internet Gateway

5. Associate the route table with the public subnet.





Step 4 – Create Security Groups

Create separate security groups for EC2 and RDS.

EC2 Security Group

Allow only the required application traffic.

For example:

SSH → Your IP only

HTTP → Required source

HTTPS → Required source

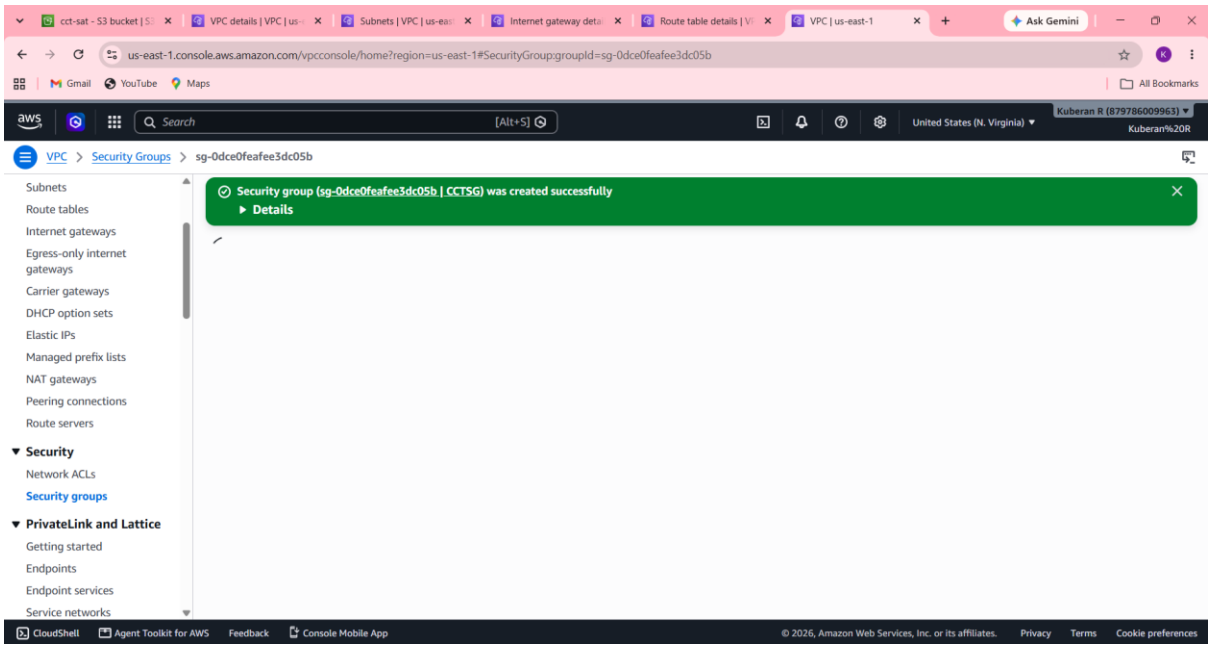
RDS Security Group

Do **not** allow database access from the entire internet.

Instead:

MySQL 3306 → EC2 Security Group

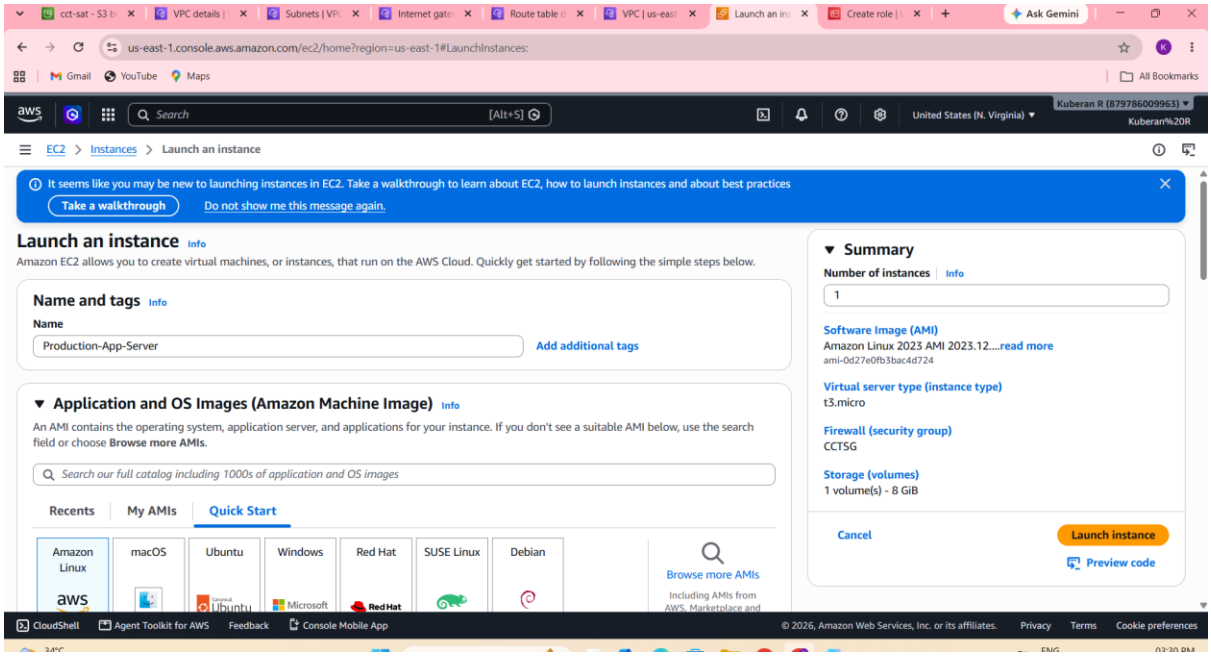
This means only the EC2 application server can communicate with the database.



Step 5 – Launch the EC2 Instance

1. Open **EC2 → Instances**.
2. Select **Launch Instance**.
3. Give it a name such as:

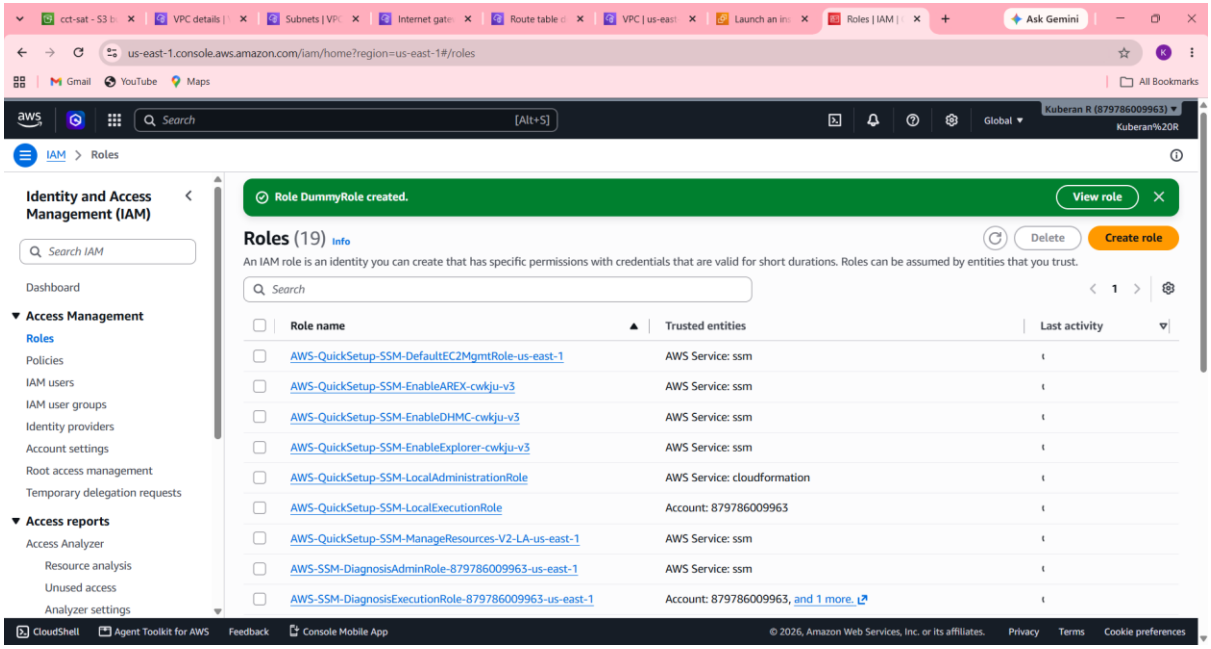
   Production-App-Server
4. Select an appropriate free-tier-eligible AMI/instance type if available to you.
5. Select the previously created VPC.
6. Select the **public subnet**.
7. Enable a public IP only if required for your implementation.
8. Select the EC2 security group created earlier.
9. Launch the instance.



Step 6 – Create an IAM Role for EC2

Instead of storing AWS access keys inside the EC2 machine:

1. Open **IAM → Roles**.
2. Create a role for **EC2**.
3. Attach a restricted S3 policy.
4. Allow only the required S3 actions.
5. Attach the role to the EC2 instance.



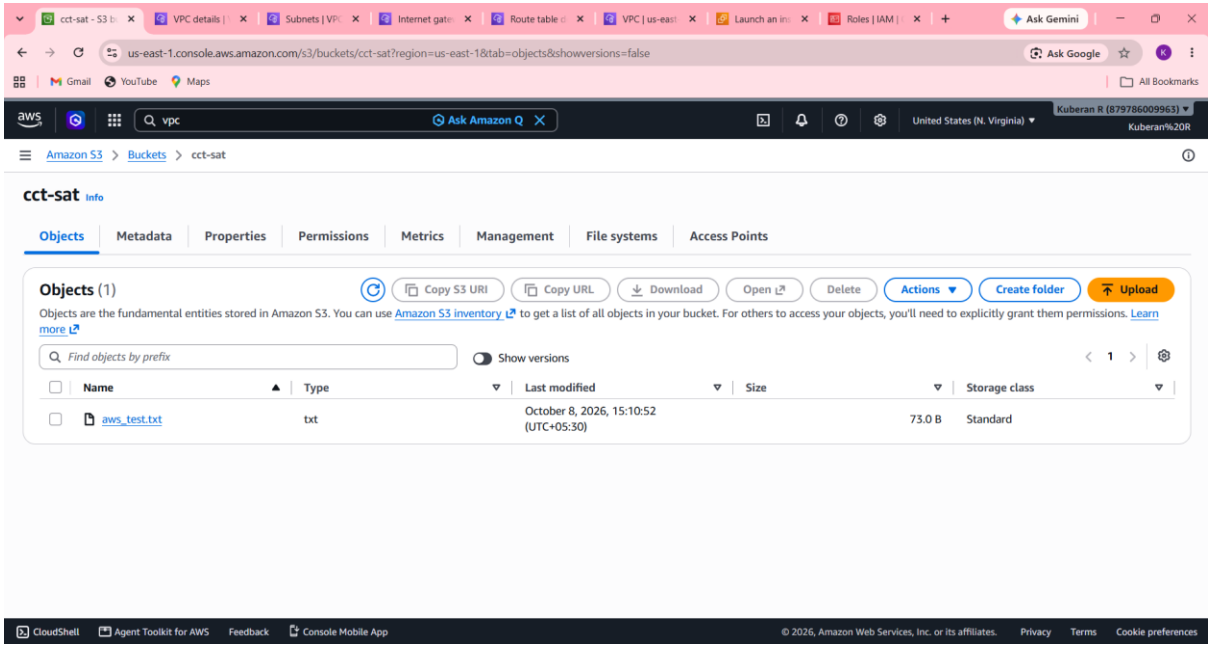
ROUND 3 – DATA INTEGRATION AND DEPLOYMENT

Objective

To integrate application data with AWS services and deploy the application using **EC2, RDS, and S3**, while validating secure communication, data access, monitoring, and deployment.

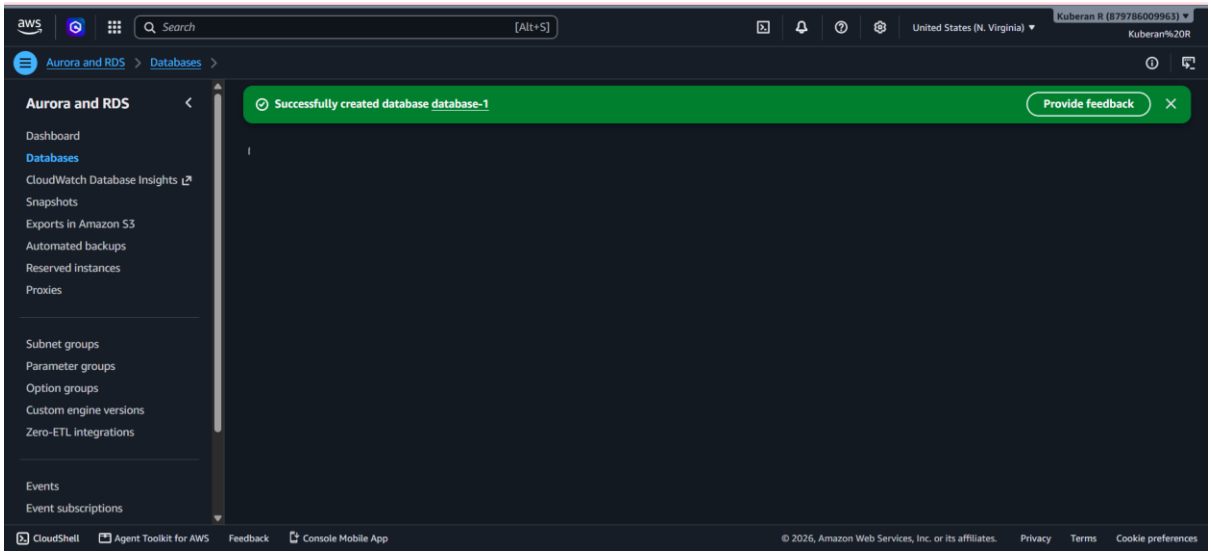
Step 1 – Upload Application Data to S3

1. Open the S3 bucket created in Round 2.
2. Create a suitable folder/prefix for application data if required.
3. Upload the required dataset/files.
4. Verify that the objects are encrypted and versioned.



Step 2 – Configure the RDS Database

1. Open the RDS database created in Round 2.
2. Connect to the database from the EC2 instance.
3. Create the required database/table structure.
4. Insert sample application data.
5. Verify that the data can be retrieved successfully.


Search
[Alt+S]
United States (N. Virginia)
Kuberan R (879786009963)
Kuberan%20R
Aurora and RDS
Databases
Aurora and RDS
Dashboard
Databases
CloudWatch Database Insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies
Subnet groups
Parameter groups
Option groups
Custom engine versions
Zero-ETL integrations
Events
Event subscriptions
Successfully created database database-1
Provide feedback
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences